# Bond Pruning: The Cross-Scale Depletion Mechanism from Atoms to Organisms

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2026


## Abstract

Every bond that forms kills another bond. The energy to close one bond is the energy released by opening another. The photon emitted at the closure carries the difference across the constraint surface at $c$ and, when absorbed, funds the next bond event elsewhere. This mechanism — formation funded by destruction, accounting carried by photons, total balanced to zero — operates identically at the atomic, neural, developmental, and organismal scales. The paper identifies bond pruning as the universal depletion mechanism of the constraint surface and proposes measurable predictions at each scale.


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## 1. The Mechanism

### 1.1 One Bond Forms, One Bond Dies

Every chemical bond formation requires energy input. That energy comes from another bond breaking somewhere in the system. The photon released at the event is not waste — it is the accounting entry, the constraint surface's receipt for a cell changing value.

**The cycle:**

1. Bond A forms $\rightarrow$ energy committed into the bond gap (0.69 Å for Zn-N)
2. Bond B dies $\rightarrow$ energy released to fund A
3. Photon emitted $\rightarrow$ carries the difference at $c$
4. Photon absorbed elsewhere $\rightarrow$ funds bond C or breaks bond D
5. Total: zero. $\omega_0 = 0$ is maintained at every step.

### 1.2 Free Radicals as Open Cells

An atom with a dangling half-bond (O•, OH•, superoxide $O_2^-$•) is a cell on the spreadsheet with an unresolved formula. The constraint surface does not tolerate open cells. Free radicals are violently reactive because the surface forces closure — the half-bond MUST find a partner or be quenched. Every antioxidant reaction is the spreadsheet resolving an open cell.

**Research direction:** Measure the time-to-closure for common free radicals (hydroxyl radical $\sim 10^{-9}$ s, superoxide $\sim 10^{-6}$ s with SOD). Does the closure time correlate with the bond gap geometry of the radical? Is the reactivity a function of the unresolved half-bond length?

### 1.3 The Bond as the Cell

Each cell on the edgeless spreadsheet is not an element. It is a bond. The periodic table lists the ATOMS — the nodes. The spreadsheet's cells are the EDGES — the bonds between them. The photon is the value propagating between cells. The periodic table balances because there is no net energy: every bond made is a bond killed elsewhere, and the photon carries the accounting.

This reframes the spreadsheet metaphor from "118 cells, one per element" to "$N$ cells, one per bond in the universe." The periodic table is the index. The bonds are the data.

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## 2. The Pruning Arc

### 2.1 Overproduction

At every scale, the system begins by overproducing:

| Scale | Overproduction | Quantity |
|---|---|---|
| Atomic (stellar) | Nucleosynthesis produces all elements to Fe-56 | 26 elements by fusion, ~90 by neutron capture |
| Molecular (prebiotic) | Random polymerisation produces vast molecular diversity | ~$10^6$ distinct small molecules in Miller-Urey experiments |
| Cellular (embryonic) | Neurogenesis produces excess neurons | ~100 billion neurons, ~$10^{15}$ synapses |
| Neural (childhood) | Synaptogenesis peaks at ~2 years | Synaptic density: ~15,000/neuron in temporal cortex |
| Organismal | Admissible set $A_0$ at maximum | Full developmental programme available |

### 2.2 Pruning

~50% of synapses are eliminated between age 2 and adolescence (Huttenlocher, 1979; Petanjek et al., 2011). The mechanism: synapses that fire together wire together (Hebb, 1949). Synapses that don't are vetoed — the constraint window closes. Each pruned synapse releases metabolic energy. That energy funds the myelination and strengthening of survivors.

**The veto is the same operator:**

- Atomic: antibonding orbital is vetoed by energy (zinc paper, §3.7)
- Neural: unused synapse is pruned by activity-dependent competition
- Developmental: unexpressed gene pathway is methylated — epigenetically vetoed
- Immune: non-reactive T-cells are eliminated in thymic selection (~95% vetoed)

### 2.3 Maturity (Plateau)

The surviving network is efficient. Fewer bonds, but each bond stronger and better myelinated. The artifact rate $r(t)$ is optimised — the monk in the mountains, burning slowly, writing fewer but more precise artifacts to the surface.

**Measurable prediction:** The metabolic rate per unit cognitive output should be LOWER in mature brains than in developing brains. The same thought costs less energy because the pruning has eliminated the redundant pathways.

### 2.4 Decline

When repair rate $\rho_{\text{prevention}}$ drops below the damage rate $\lambda_{\text{attempt}}$:

- Bonds break faster than they form
- Synapses prune without replacement
- The admissible set shrinks
- The depletion curve steepens

The pruning that was adaptive in childhood becomes pathological. Not because the mechanism changed — because the budget has been spent. The same operator, the same veto, the same constraint window closing. Only now it closes on cells the system still needs.

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## 3. Cross-Scale Measurements

### 3.1 The Same Curve

**Hypothesis:** The overproduction → pruning → plateau → decline curve has the same mathematical form at every scale. The prediction is a universal depletion function:

$$N(t) = N_0 \cdot f\left(\frac{t}{\tau}\right)$$

where $N_0$ is the initial count (bonds, synapses, cells), $\tau$ is the scale-specific time constant, and $f$ is the universal shape function — rising to a peak, pruning to a plateau, then declining.

**Testable at:**

| Scale | Observable | Measurement method | Expected curve shape |
|---|---|---|---|
| Chemical bonds | Bond count in a reaction network | Calorimetry (enthalpy tracking) | Rise → plateau → fall |
| Synapses | Synaptic density | Post-mortem histology, PET imaging (SV2A) | Rise → prune → plateau → decline |
| White matter | Myelination | MRI diffusion tensor imaging | Rise → plateau → decline |
| Immune cells | T-cell diversity | Flow cytometry, TCR sequencing | Rise → thymic pruning → plateau → decline |
| Metabolic rate | Basal metabolic rate per kg | Indirect calorimetry | Rise → plateau → decline |
| Organismal | $\lvert A(t)\rvert$ (admissible set proxy) | Composite of above | Rise → prune → plateau → decline |

### 3.2 The Photon Budget

If every bond that forms kills another, and every bond event emits a photon, then the total photon output of an organism over its lifetime is a direct measure of its bond turnover — its artifact budget.

**Prediction:** Total metabolic heat output over a lifetime (integral of basal metabolic rate) should correlate with $E_0$ — the initial energy bonded at the zinc spark. Kleiber's law ($P = P_0 M^{3/4}$) already shows that metabolic rate scales with mass. The bond-pruning framework predicts that

lifetime metabolic output ($P \times t_d$) is approximately constant per unit mass across species — the same budget, different spending rates.

**Literature to check:** Rubner's "rate of living" hypothesis (1908). Speakman (2005) challenged it. Does the constraint-surface framing resolve the discrepancy?

### 3.3 Free Radical Closure Times

**Prediction:** Free radical reactivity (rate constant $k$) should correlate with the unresolved bond gap — the geometric distance between the radical's electron density and the nearest available bonding partner. Radicals with larger unresolved gaps should be MORE reactive (the spreadsheet forces closure harder on bigger gaps).

| Radical | Half-life | Bond gap (predicted) | Reactivity |
|---|---|---|---|
| OH• (hydroxyl) | ~$10^{-9}$ s | Large — single unpaired electron, no steric shielding | Highest |
| $O_2^-$• (superoxide) | ~$10^{-6}$ s (with SOD) | Medium — mesomeric stabilisation | High |
| NO• (nitric oxide) | ~1 s in tissue | Small — stable radical, partially delocalised | Moderate |
| Vitamin E radical | ~seconds | Very small — extensive conjugation stabilises | Low |

### 3.4 Neural Pruning Rate vs Bond Energy

**Prediction:** The rate of synaptic pruning in a cortical region should correlate with the metabolic activity of that region during the pruning window. Higher activity → more bond turnover → faster pruning of unused connections. Prefrontal cortex (last to myelinate, last to prune) should show the longest pruning window and the highest cumulative metabolic cost of pruning.

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## 4. The Conservation Law

### 4.1 Bond Accounting

Every bond event on the constraint surface satisfies:

$$\Delta E_{\text{formed}} + \Delta E_{\text{broken}} + E_\gamma = 0$$

where $\Delta E_{\text{formed}}$ is the energy committed to the new bond, $\Delta E_{\text{broken}}$ is the energy released by the broken bond, and $E_\gamma$ is the photon energy. The photon carries the residual — the accounting mismatch between what was broken and what was formed.

This is $\omega_0 = 0$ at the bond level. Every bond event balances to zero. The constraint surface is closed under bond operations, just as the periodic table is closed under nuclear operations (One Skin, §1.4).

### 4.2 Why Bonds Only Grow One Way

**Building thought.** The bond length measured at the moment of closure is a snapshot. The "half-bond" is not a property each atom carries — it is the measurement at the instant of connection. Before the bond closes, each atom is reaching. The reaching is directional — from higher energy to lower energy, from less stable to more stable. Bonds grow TOWARD the fixed point, never away from it.

This is why bond formation is exothermic for stable configurations and endothermic for unstable ones. The constraint surface pulls bonds toward the minimum energy configuration. The direction is set by $\alpha$ — the coupling constant determines which direction is "downhill." The bond does not choose. The gradient chooses. The bond is the gradient made physical.

Each connection changes everything at the atomic bond level. Every atom in the vicinity of a new bond has its electron density redistributed. The polarisation propagates outward — a chain reaction of subtle adjustments, each atom re-reaching toward its neighbours with slightly different half-bond extensions. The new bond doesn't just connect two atoms. It recalculates every cell in the local neighbourhood of the spreadsheet.

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## 5. Boundaries

1. **The "one bond kills another" claim is conservation, not mechanism.** The energy balance is guaranteed by the first law of thermodynamics. The claim is that this conservation operates as a bond-level veto — each formation IS a destruction elsewhere — not that specific bond pairs are linked. The pairing is statistical, not deterministic.
2. **The universal depletion curve is a hypothesis.** The prediction that the same $f(t/\tau)$ shape function appears at atomic, neural, immune, and organismal scales requires cross-scale data that has not been assembled. Each scale has its own literature. The synthesis is the contribution; the individual data are published.
3. **Free radical closure times are measured, not predicted.** The framework reinterprets known rate constants as bond-gap-driven closure events. It does not predict new rate constants from first principles. The measurable prediction is the correlation between gap geometry and reactivity, not the absolute values.
4. **Synaptic pruning literature is established.** Huttenlocher (1979), Petanjek et al. (2011), and Rakic et al. (1986) documented the overproduction-pruning arc decades ago. The contribution is not the pruning data — it is the identification of pruning as the same veto operator that governs bond formation, death, and nuclear decay.

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## References


**Internal (Carpenter, 2026)**

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- (2026e) "The Constitutional Constant." DOI: 10.5281/zenodo.18995286
- (2026f) "The Möbius Strip as Fixed Point." DOI: 10.5281/zenodo.19020526
- (2026q) "Tetrahedral Zinc." DOI: 10.5281/zenodo.19032617
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